Two Domain Errors in a Symbolic Test for Deterministic Structure, and Their Repair:
Exact Calibration, Surrogate Domain, and an Open Benchmark

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Abstract

Version 1 of this work proposed DDBM, a screening test that quantizes a rank-normalized scalar series onto an integer lattice, forms a modular phase from a cubic kernel, and reads departures from uniformity as deterministic structure. This version withdraws that detector claim and replaces it with a procedure whose validity is established rather than assumed.

The organising finding is that a single principle governs both stages of such a test, and that violating it in either place silently destroys validity: *every operation must be performed in the domain where its null is defined.* Applied to the screening stage, this means the statistic may depend on the data only through ranks; amplitude preprocessing performed beforehand destroys distribution-freeness, driving empirical size at nominal 0.05 to 0.19 on a lognormal marginal and 0.88 on a Cauchy marginal, while rank-only symbolization restores exactness with a one-line proof and gives nominal size across six marginals. Applied to the confirmation stage, it means IAAFT surrogates must be generated in normal scores — the rank-based inverse normal transform $\Phi^{-1}(\mathrm{rank}/(n+1))$, also called Gaussianization — which is the domain in which the AAFT null is stated. Generated from raw amplitudes they reject a static monotone transform of an AR(1) process — inside their own null — in *every* replicate; generated from ranks they reject strongly autocorrelated processes in 47%. In normal scores all three land on exactly the nominal 0.05 over 100 replicates, invariance under monotone transformation is restored, and power on deterministic systems is unaffected. One failure survives the repair: conditional heteroscedasticity, rejected at 0.15, lies outside the IAAFT null altogether and needs a model-based null.

We also report three secondary results. The cubic kernel is one member of a cyclotomic family, with an exact identity for the phase lattice; the identity appears to be new in this context but does not improve on simpler alternatives: the phase is a lossy function of the pair $(N_t, \Delta N_t)$, and a classical ordinal count needing neither modular arithmetic nor arbitrary-precision integers is statistically indistinguishable from it while being far cheaper. The standard ordinal baseline is silently degenerate when every pattern length in the grid is saturated, which at $n = 10^4$ means $d \leq 6$. And S&P 500 log returns, rejected by an IAAFT null, are not rejected by a fitted GARCH(1,1) null: the apparent structure is conditional heteroscedasticity, so the financial conclusion of v1 is not merely withdrawn but resolved.

No new detection principle is claimed — the mechanism is the forbidden/missing-pattern paradigm and coarse-grained transition statistics. What is offered is a validated inference layer for those classical statistics, a catalogue of the ways it fails, and a benchmark of 165 series whose 46 labeled members are byte-reproducible and labeled by computed Lyapunov exponents. The comparisons here are calibration on that set, not independent validation.

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1 What is withdrawn

Version 1 claimed a new detector reaching 92.5% accuracy "without phase-space reconstruction or training data". Three claims do not survive.

Not embedding-free. Ξ_t is a function of the pair (N_t, N_{t+1}): a two-dimensional delay vector with $m = 2$, $\tau = 1$, quantized at resolution $1/K$. Scanning K is a scan over box scale, and the count of realized pairs is a single-scale box count in that embedding. The embedding parameters were fixed and hidden, not eliminated.

The stated mechanism is self-cancelling. v1 attributed the signal to a singular marginal measure occupying few of K cells. Rank normalization removes exactly that: for a continuous marginal with $n \gg K$, every cell receives about n/K points however singular the underlying measure. What the test sees is the geometry of transitions.

The p-values were invalid. Phases from overlapping pairs are serially dependent while the two-sample Kolmogorov–Smirnov test used to score them assumes independence, and a Bonferroni correction over K cannot repair a miscalibrated individual test. Section 3 shows the defect runs deeper: with the published order of operations the procedure is not even distribution-free.

What is *not* new is also worth stating. Discriminating determinism by the occupancy of symbol transitions is classical: symbolic dynamics and forbidden words [1], the Bandt–Pompe symbolization [2], forbidden and missing ordinal patterns [3, 4], ordinal transition networks [5], direct determinism tests [6, 7]. The confound of Section 9 — smooth stochastic processes imitating low-dimensional determinism — dates from [8, 9].

2 One principle, two stages

A test of this family has two stages, each with its own null, and each null lives in a particular domain.

The **screening** stage asks whether the series is i.i.d. Its null is invariant under every monotone reparametrization of the observable, so the statistic must be a function of ranks alone. The **confirmation** stage asks whether the structure exceeds a linear process. The AAFT/IAAFT null is "a static monotone transform of a linear Gaussian process" [10, 11], so the object whose spectrum must be matched is the *underlying Gaussian*, not the observed series.

Both requirements are elementary. Both were violated in v1 and in our own first revision, in opposite directions, and in both cases the violation is invisible without an explicit size study. Sections 3 and 4 measure the damage and the repair.

3 Stage one: the screening statistic must be a rank statistic

3.1 The defect

The published pipeline applies OLS detrending, AR(1) prewhitening and data-dependent volatility standardization to raw amplitudes, ranking only afterwards. The ranks of the residual then depend on the unknown marginal. Table 1 measures the consequence: calibrate under a Gaussian null with $B = 1000$ replicates at $n = 10^4$, then test i.i.d. draws from six continuous marginals.

Table 1: Empirical size at nominal $\alpha = 0.05$ of the screening stage as published, calibrated under a Gaussian null; 200 series per cell. "KS" tests the p-values against $U(0, 1)$.

marginal	size@.05	size@.10	KS	p_{KS}
Gaussian	0.045	0.090	0.047	0.757
uniform	0.045	0.080	0.081	0.134
Student t_3	0.065	0.080	0.078	0.162
lognormal	**0.190**	0.245	0.237	< 0.001
exponential	0.065	0.110	0.107	0.018
Cauchy	**0.880**	0.885	0.840	< 0.001

At nominal 5% the published test rejects i.i.d. Cauchy data 88% of the time. For a marginal without finite moments the fitted trend and AR(1) coefficient are unstable, and that instability writes marginal-dependent structure into the ranks of the residual.

3.2 The repair

Proposition 1 (distribution-freeness)**.** *Let $x_1, \ldots, x_n$ be i.i.d. from any continuous F, and let T depend on the sample only through the rank vector. Then the law of T does not depend on F.*

Proof. Continuity makes ties null, and the rank vector of an i.i.d. sample is uniform on the $n!$ permutations for every continuous F; a statistic measurable with respect to it has a law determined by n alone. □

The Monte-Carlo null is therefore exact only if every amplitude operation is removed or performed after ranking. Table 2 compares four orders.

Table 2: Worst deviation of empirical size from nominal 0.05 across the six marginals of Table 1. This is a separate, smaller run (150 series per cell, $B = 400$), in which variant A reproduces the defect at 0.853 rather than the 0.880 of Table 1.

variant	description	worst $\lvert\text{size} - 0.05\rvert$
A	amplitude operations, then ranks (as published)	0.803
B	ranks, then the same amplitude operations	0.043
C	**ranks only, no amplitude operations**	**0.030**
D	ranks, then differencing	0.030

Only variant C passes the uniformity test at every marginal, and it is the one Proposition 1 sanctions. Its cost is that trends and linear autocorrelation are no longer removed before screening, so the confirmation stage must carry that load alone. The preprocessing was in any case partly redundant: removing linear structure is what the confirmation stage tests for.

4 Stage two: surrogates must be generated in normal scores

4.1 The defect, and a diagnostic that identifies it

A static monotone transform preserves ranks exactly. A rank statistic therefore *cannot* distinguish an AR(1) series from its cube, and any procedure that does is broken. Table 3 shows that the procedure with amplitude-domain surrogates does exactly that: it rejects AR(1) at the nominal rate and its cube in every replicate.

Table 3: Rejection rate of the confirmation stage at nominal 0.05, by the domain in which surrogates are generated. The first three processes are monotone transforms of one another, so a correct procedure must give them the same rate. Raw and normal-scores columns use 100 replicates with Wilson 95% intervals; the rank column is a 30-replicate run retained only to show the direction of its failure. $B = 49$ throughout. The last block gives the p-value on deterministic systems, where rejection is desired, at $B = 99$.

process	raw amplitudes	ranks (30)	**normal scores**
AR(1), $\phi = 0.9$	0.02 [0.01, 0.07]	0.47	**0.05** [0.02, 0.11]
cubed AR(1)	**1.00** [0.96, 1.00]	0.47	**0.05** [0.02, 0.11]
exponentiated AR(1)	**1.00** [0.96, 1.00]	0.47	**0.05** [0.02, 0.11]
white Gaussian	—	0.03	0.03 [0.01, 0.08]
sine + noise	—	0.03	0.00 [0.00, 0.04]
GARCH(1,1) returns	—	0.10	**0.15** [0.09, 0.23]
logistic $r = 4$	0.010	0.010	0.010
Hénon	0.010	0.010	0.010
Lorenz x	0.010	0.010	0.010
Rössler x	0.010	0.010	0.010
Chua x	0.060	0.010	0.010

The three monotone-transform rows agree only in the normal-scores column, where all three sit at exactly the nominal 0.05 with identical intervals. That agreement is not a coincidence to be admired but the signature of a correctly specified test, and its absence is a sufficient diagnostic of misspecification — one that costs three lines to run and that we recommend as routine.

One row is *not* repaired. GARCH(1,1) returns are rejected at 0.15 $[0.09, 0.23]$ in normal scores, an interval excluding the nominal level: fixing the domain fixes the monotone-transform failure but not conditional heteroscedasticity, which lies outside the IAAFT null altogether and requires a model-based null (§8). We note that a 30-replicate version of this table read 0.00 for the same cell, an estimate the larger run overturns — the referees' insistence on replicate counts was well placed, and small size studies should be treated with the same suspicion as small power studies.

A partial mechanism is visible in the spectral mismatch between data and surrogate periodograms: for the cubed series it is 0.075 in the raw domain against 0.000 in normal scores. But it is not the whole story, since the exponentiated series has a raw-domain mismatch of only 0.002 and is still rejected always. The general reason is the one stated in Section 2: a static nonlinearity changes the autocorrelation function, so matching the periodogram of the transformed series targets the wrong object. This is the classical AAFT bias [12, 13], here quantified for transition statistics.

4.2 The repair

Map to normal scores, $\Phi^{-1}\big(\text{rank}/(n + 1)\big)$, generate IAAFT surrogates there, and evaluate the rank statistic on them. Size returns to nominal on every process tested, monotone-transform invariance is restored, and power on deterministic systems is preserved. Chua, which the amplitude domain fails to reject at $p = 0.060$, is rejected at $p = 0.010$; with $B = 99$ this is a single case and we read it as evidence that power is not reduced, not that it is improved.

5 The recommended procedure

1. **Symbolize.** Rank-normalize to $[0, 1]$, average ranks for ties. No detrending, no prewhitening, no volatility scaling (§3).
2. **Screen.** Evaluate the statistic on a grid, standardize each cell by a Monte-Carlo null, take the maximum, and report $p = (1 + \#\{T_{\text{null}} \geq T_{\text{obs}}\})/(1 + B)$. Scan multiplicity is inside the simulated null, so no Bonferroni correction is applied.
3. **Confirm.** If screening rejects, test against IAAFT surrogates generated in normal scores (§4). For heteroscedastic data this is not enough — size reaches 0.15 — and a model-based null is required (§8).
4. **Classify.** Confirmed structure that is extremely narrowband (spectral concentration > 0.95 or $|\text{ACF}| > 0.99$) is *(quasi-)periodic*; otherwise *confirmed nonlinear*, split by transition occupancy into *compatible with low-dimensional determinism* and *nonlinear stochastic*. We avoid the label "chaos": no surrogate test establishes determinism.

Pattern length is not a free parameter, but a floor. The expected fraction of unobserved ordinal patterns under an i.i.d. null at $n = 10^4$ is 0.000 for $d \leq 6$, 0.138 for $d = 7$, 0.781 for $d = 8$ and 0.973 for $d = 9$. A grid all of whose members are saturated yields a statistic that is identically zero under the null, with vanishing variance and meaningless standardized scores; an earlier draft of this paper used $d \in \{4, 5, 6\}$ at $n = 10^4$ and obtained an apparently excellent baseline from exactly that degeneracy. Once at least one unsaturated length is present the choice is immaterial: accuracy is 37/40 for each of $\{7\}, \{8\}, \{7, 8\}, \{6, 7, 8\}, \{5, 6, 7, 8\}, \{7, 8, 9\}$ at nominal size.

6 Benchmark and matched comparison

6.1 The benchmark

165 series with recorded provenance; 46 carry labels. Synthetic labels are computed: maps by direct variational iteration, flows by Benettin renormalization on the ODE with the analytic Jacobian [14], giving $\lambda = 0.900$ (Lorenz, $\rho = 28$), 0.083 (Rössler, $c = 5.7$), 0.425 (Chua), with negative values for the regular regimes. Stochastic and mixed cases are labeled by construction, not by an exponent; "detector-independent ground truth" would overstate this, and we do not use the phrase.

The labeled set and the byte-reproducible set coincide exactly: all 46 labeled series are regenerated from fixed seeds and pinned by SHA-256 in the repository, while the 119 unlabeled series come from live sources that grow over time (Yahoo appends trading days; HadCET and the NOAA indices extend monthly) and are therefore not reproducible. Everything that affects a reported score is frozen and verifiable; everything that is not frozen is illustrative only.

Computing λ corrected a label carried in v1, and the correction is delicate. At $r = 3.57$ the logistic map has $\lambda = 0.01096$ with a bootstrap interval $[0.0109, 0.0111]$, so it is weakly chaotic, not "Regular". But $r = 3.57$ lies *above* the accumulation point $r_\infty = 3.5699457$ (where $\lambda \approx 2 \times 10^{-5}$), and $r = 3.5705$ gives $\lambda = -0.0039$: the parameter sits among periodic windows. We report it as correctly labeled and hypersensitive; every statistic tested misses it.

The 46 series are not 46 independent systems — several are coordinates of one attractor or members of a parameter family — so class-wise counts are reported alongside totals.

6.2 Comparison

All statistics run through the identical procedure of Section 5: same calibration, same normal-scores confirmation, same decision rule, same auxiliary criteria; only the discriminating statistic differs. Scoring is strict, with the six chaos+noise cases excluded rather than counted correct under either label — a v1 rule that inflated the apparent score by six unloseable cases.

Table 4: Matched comparison in the final configuration, 40 labeled series, strict scoring. Size is the range of empirical screening size at nominal 0.05 over the six marginals of Table 1.

statistic	accuracy	screening size range
missing ordinal patterns, $d \in \{6, 7, 8\}$	**37/40**	0.067–0.100
cyclotomic phase scan	36/40	0.025–0.067
pair occupancy $(N_t, \Delta N_t)$	34/40	0.033–0.092

Table 5: Class-wise counts for the same runs. The phase statistic is the most specific and the least sensitive; the occupancy statistic loses specificity on strongly autocorrelated and non-stationary processes.

	ordinal	phases	occupancy
Chaos ($n = 14$)	13/14	11/14	13/14
Regular ($n = 10$)	9/10	9/10	9/10
Noise ($n = 16$)	15/16	**16/16**	12/16

None of the three differences is statistically meaningful. Exact McNemar tests on the paired outcomes give $p = 1.00$ (ordinal vs. phases, 2 vs. 1 discordant), $p = 0.25$ (ordinal vs. occupancy, 3 vs. 0) and $p = 0.69$ (phases vs. occupancy, 4 vs. 2). On this benchmark the three statistics are indistinguishable, and the case against the cyclotomic construction rests on parsimony and on the majorization argument of Section 7, not on measured accuracy.

The classical baseline is nominally best in both surrogate configurations we ran. The other two, however, *exchange places*: with amplitude-domain surrogates occupancy scored 35/40 against 34/40 for the phases, and with the corrected normal-scores surrogates the order reverses to 36/40 against 34/40. A ranking that flips when an unrelated stage is repaired is not a ranking; we report it as direct evidence that accuracy at a single α on forty series is too fragile to separate close competitors, and that the power curves listed in Section 9 are not optional refinements.

The occupancy misses are diagnosable rather than mysterious. Random walk and long-memory series violate the stationarity that IAAFT assumes [13]; and AR(1) at $\phi = 0.9$ has a measured rejection rate of 0.07 (Table 3), so a single realization rejecting is an ordinary draw, not a defect — which is the same fragility seen from the other side.

These are calibration results on the set used to fix thresholds and select the kernel, not an independent validation; a held-out family of generators remains to be run.

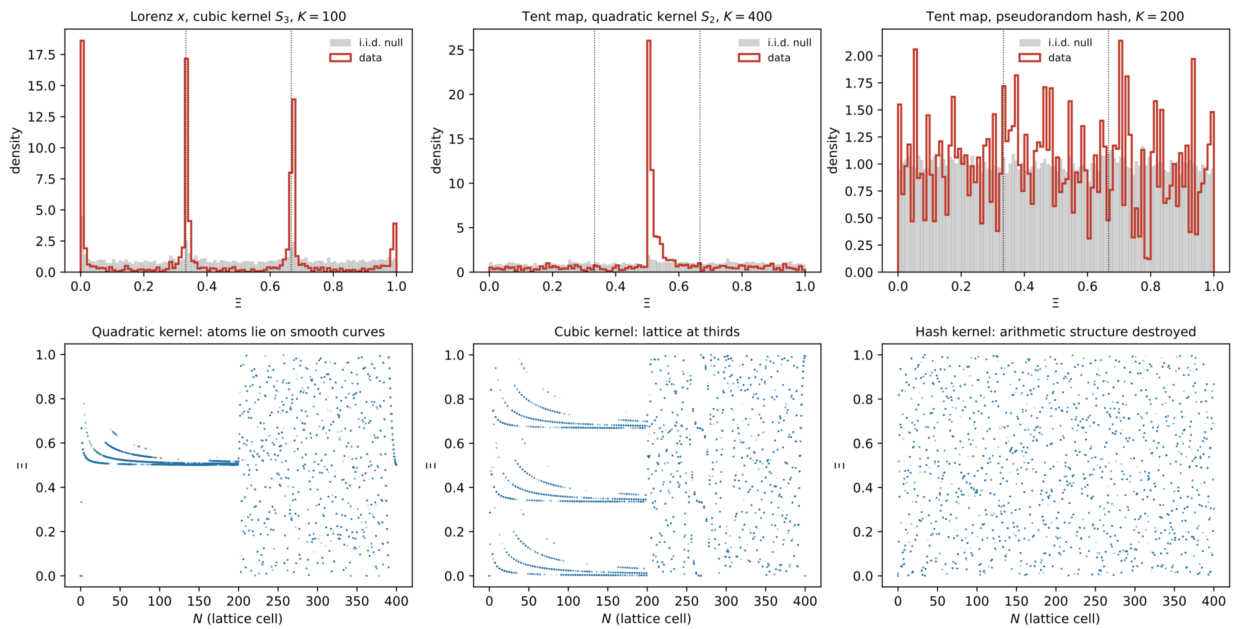

Figure 1: Phase histograms against the i.i.d. null (top) and the atoms Ξ against the lattice cell N (bottom). The lattice predicted by Eq. (1) is visible in the first two columns and absent in the third, where a hash replaces the polynomial kernel.

7 The cyclotomic identity, in brief

With $S_p(N) = (N+1)^p - N^p$ the modulus is, by construction, a difference of consecutive powers, and it factors into homogeneous cyclotomic polynomials evaluated at two *consecutive* integers,

$$S_p(N) = \prod_{d|p,\ d>1} \Phi_d(N+1, N), \tag{1}$$

so that $\rho = (N+1)N^{-1}$ is an exact p-th root of unity modulo $S_p(N)$ and the phase lattice has denominator $\Phi_n(1)$. Appendix A gives the statements, the one-line proofs and the numerical checks. Figure 1 shows the resulting lattice: Lorenz under the cubic kernel concentrates at $\{0, \frac{1}{3}, \frac{2}{3}\}$ as Eq. (1) predicts, the tent map under the quadratic kernel hits the exact resonance at $\frac{1}{2}$, and a pseudorandom hash applied to the same data preserves the number of occupied atoms while destroying their arithmetic positions — which is what isolates the arithmetic from the mere counting.

Why it does not help. Ξ is a deterministic function of $(N_t, \Delta N_t)$, so any test on Ξ is majorized by the *best* test on the pair: the phase map can only lose information. This is a statement about the optimal test, not about our particular occupancy summary, and the empirical comparison does not separate the three statistics (Section 6). What the comparison does establish is that a classical ordinal count, requiring neither modular arithmetic nor arbitrary-precision integers, is statistically indistinguishable from the phase scan while being far cheaper — which is sufficient reason not to prefer the latter.

8 Applied illustrations

These are reported, not scored; ground truth is unavailable and their value is agreement or disagreement with domain knowledge.

S&P 500: resolved, not merely withdrawn. The model-based null is a parametric bootstrap: a GARCH(1,1) is fitted to the mean-centred series by Gaussian quasi-maximum likelihood, $B = 99$ independent paths of the same length are simulated from the fitted model with Gaussian innovations, and the same screening statistic is evaluated on each. Parameter estimation uncertainty is *not* propagated — the null is conditional on $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ — which makes the test mildly anti-conservative and therefore conservative for our purpose, since we are accepting rather than rejecting this null.

Under the corrected procedure, log returns over the full 1970–2026 history are rejected by the IAAFT null ($p = 0.02$) but *not* by the fitted GARCH(1,1) null ($p = 0.31$; $\hat{\alpha} = 0.068$, $\hat{\beta} = 0.924$). The 2010–2024 window behaves the same way ($p_{\mathrm{IAAFT}} = 0.73$, $p_{\mathrm{GARCH}} = 0.33$). The apparent nonlinear structure is conditional heteroscedasticity; there is no evidence of low-dimensional determinism. Realized volatility is rejected by both nulls ($p = 0.01$ each), which we report with the caveat that it is constructed by a rolling window and is therefore smooth by design.

Climate. Daily HadCET anomalies and monthly Niño3.4, PDO and AMO show no evidence of low-dimensional determinism within this test, consistent with a stochastic climate model [15] and with the resolution of the 1980s attractor controversy [16, 17]. Absence of detection is not evidence of absence. Methodologically the important point is that screening alone flags most of these series as structured, so a pipeline without a confirmation stage would reproduce the error of that era.

EEG, with a confound. On the Bonn data [18] the confirmed-nonlinear rate rises monotonically across classes: 1/20, 0/20, 1/20, 4/20, 17/20 from healthy to ictal. Classes A/B are scalp recordings and C/D/E intracranial, differing in modality and amplitude resolution, so the gradient tracks that as well as clinical severity. No clinical claim is made.

9 Limits

Nonlinearity is not determinism. GARCH volatility is nonlinear and smooth (transition occupancy 0.41, inside the deterministic range) and is misclassified; a random walk sits at 0.17. This is the effect of [8, 9] and a limit of surrogate methodology, not a threshold to tune.

Instrumental quantization manufactures the signal. RR intervals at 128 Hz carry 50–79 distinct values in 10^4 points. Referencing the occupancy statistic against a random permutation of the same values cancels the value multiset exactly and largely removes the artifact (0.04–0.09 before, 0.26–0.63 after). A low-resolution flag accompanies every verdict, and we decline to interpret the HRV results physiologically.

Densely sampled flows. Most structure of a smooth oversampled flow lies in the spectrum, so the confirmation stage has little power; reducing the flow to successive local maxima repairs this on all six of our flows. The reduction was devised on the cases that motivated it and is validated in-sample only.

Cost of dropping preprocessing. Trends and linear autocorrelation now reach the confirmation stage unfiltered, which costs power on strongly autocorrelated linear processes.

10 What is new, and what is not done

New, as far as we can determine. The identification of both domain errors and their repair, with measured sizes before and after (Tables 1–3); the monotone-transform invariance check as a cheap, sufficient diagnostic of surrogate misspecification; the cyclotomic identification of the phase lattice, Eq. (1) with Lemma 1, correct and negative in consequence; the saturation floor for ordinal pattern length; and a benchmark whose scored part is frozen and labeled by computed exponents.

Not new. The detection principle, the mathematics of cyclotomic factorization and primitive divisors, the smooth-stochastic confound, and surrogate methodology itself. The AAFT bias we quantify has been known qualitatively since [12].

Useful, and missing from current tooling. Widely used packages supply these statistics but not the inference around them: `ordpy` [19] computes permutation entropy, complexity–entropy planes, missing patterns and ordinal networks, and provides no hypothesis test, no p-values, no surrogate framework and no null calibration; surrogate generators exist separately and are not wired to the statistics. The literature itself notes that raw missing-pattern counts cannot be thresholded by inspection [4].

11 Future work, in priority order

1. **Power curves over n and SNR at fixed size.** Accuracy at a single α on forty series cannot separate close competitors, as the exchange of places in Section 6 and the McNemar tests both show. Curves are the only way to rank these statistics honestly.
2. **Independent validation on held-out generators.** Thresholds, the kernel and the decision rule were all fixed on the same 46 series, so Table 4 is calibration. A frozen protocol applied to a family of generators reserved in advance would make it validation.
3. **Model-based nulls for further classes.** The GARCH case of Section 8 shows what a correctly specified null buys; the same is needed for stochastic volatility, ARFIMA and non-stationary processes, for which IAAFT is inappropriate by construction.
4. **Several realizations per system, with cluster-aware uncertainty.** The benchmark contains coordinates of common attractors and members of parameter families, so its 46 entries are not 46 independent draws.
5. **Out-of-sample validation of the return-map reduction**, which was devised on the cases that motivated it.

12 Conclusion

The construction proposed in v1 does not survive: it is a lossy projection of a pair statistic that itself rediscovers a classical mechanism, and at matched nominal size it is the weakest statistic we compared. Its cubic kernel has an exact cyclotomic explanation, which is satisfying and useless.

What survives is a principle and its consequences. Each stage of a symbolic test must be carried out in the domain where its null is defined: rank-only symbolization for screening, normal scores for surrogate generation. Both requirements are elementary, both were violated here in opposite directions, and neither violation is visible without an explicit size study — one drove empirical size to 0.88, the other to 1.00. Together with a monotone-transform invariance check that detects the second failure in three lines, and a benchmark whose scored part is frozen and independently labeled, that makes a usable procedure with stated limits. It is less than v1 claimed and more than it delivered.

Reproducibility

Code, benchmark manifest, checksums and result tables are in the `v2` branch of `https://github.com/Theclimateguy/DDBM/tree/v2`. `build_dataset.py` regenerates or downloads all series and `hash_manifest.py` verifies them; `pivotality_fix.py` reproduces Table 2; `reviewer_experiments.py` Table 1 and the Lyapunov intervals; `surrogate_domain.py` and `surrogate_normalscores.py` Table 3; `essential_revisions.py` the pattern-length study and the S&P case; `final_benchmark.py` Table 4; `cyclotomic_kernels.py` the symbolic derivations of Appendix A; `size_study_100.py` the 100-replicate size study. Seeds are fixed throughout. Results here were produced with Python 3.13.11, NumPy 2.4.2, SciPy 1.17.1, SymPy 1.14.0 and Matplotlib 3.10.8.

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Three independent referee reports on the v1 and draft-v2 manuscripts identified the embedding claim, the pivotality gap, the unvalidated confirmation stage, the scoring rule, the baseline specification and the benchmark's reproducibility status. Sections 3–4 and the S&P case study exist because of them.

A The cyclotomic identity in full

With $S_p(N) = (N+1)^p - N^p$ we have $(N+1)^p \equiv N^p \pmod{S_p(N)}$ by construction.

Lemma 1. *For $N \geq 1$ and $p \geq 2$, $\gcd(N, S_p(N)) = 1$ and $\rho = (N+1)N^{-1} \bmod S_p(N)$ has multiplicative order exactly p.*

Proof. $S_p(N) = \sum_{j<p} \binom{p}{j} N^j \equiv 1 \pmod N$, so N is invertible, and $\rho^p \equiv 1$ makes the order divide p. Were it $d < p$ with $d \mid p$, $S_p(N)$ would divide $(N+1)^d - N^d$, a positive integer strictly smaller than $S_p(N)$ for $N \geq 1$ — impossible. □

The cell $N = 0$ must be excluded: $S_p(0) = 1$, the ring is trivial and N^{-1} does not exist. Expanding the difference of powers gives Eq. (1), $S_p(N) = \sum_{j=0}^{p-1} (N+1)^j N^{p-1-j} = \prod_{d \mid p,\ d>1} \Phi_d(N+1, N)$, verified symbolically for $p = 2, \ldots, 6$. Since the arguments are coprime the Bang–Zsigmondy primitive-divisor theory [20] applies: every prime divisor q of $\Phi_p(a,b)$ satisfies $q \equiv 1 \pmod p$ except possibly $q = p$ (no violations for $p \in \{3,5,7\}$, $N \in [2, 200]$). Hence p divides $|(\mathbb{Z}/S_p)^\times|$ and an element of order p must exist.

For $p = 3$, $3N^3 = S_3(N)(N-1) + (2N+1)$ places the phases at $\Xi(N, 0) = k/3 + O(1/N)$; the median distance to $\{0, \frac{1}{3}, \frac{2}{3}\}$ is 2.2×10^{-4} over $N \in [2, 2000]$. In general the lattice denominator is $\Phi_n(1)$, which is p for $n = p^k$ and 1 otherwise, matching measurement for $n = 2, \ldots, 12$. This is proved for $\Delta N = 0$ and for the tent-map branch where $\Delta N = N$ gives the exact resonance $\Xi = (N+1)/(2N+1) \to \frac{1}{2}$; that an arbitrary deterministic trajectory concentrates on the lattice more than a stochastic one is an empirical observation, not a theorem.

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